

Daniel Thomas Jesudoss

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Software engineer with an MS in Computer Science (UT Arlington). Built and deployed production systems independently — a multi-tenant RAG document platform (Java 21 / Spring Boot / React) and zero-trust Kubernetes infrastructure — with core strengths in Python, Java, and TypeScript across backend, frontend, and cloud.

Education

Master of Science, Computer Science | GPA: 3.9

The University of Texas at Arlington

May 2026

Arlington, TX

- Relevant Coursework: Design and Analysis of Algorithms, Software Engineering, Intelligent Systems

Bachelor of Technology, Information Technology

Loyola-ICAM College of Engineering and Technology

Apr 2024

Chennai, India

Technical Skills

Languages: Python, TypeScript, Java / JavaScript, SQL (PostgreSQL)

Databases: PostgreSQL, pgvector (HNSW cosine index), Flyway Migrations, AWS S3, AWS SQS

Testing: JUnit, Mockito, unit test design, integration test design

Frameworks: Spring Boot, React.js, Next.js, Flask, Node.js, LangChain4j

Cloud & DevOps: AWS (EC2, S3, SQS, Lambda, Route 53), Docker, GitHub Actions CI/CD, Kubernetes (K3s)

AI & Systems: Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Integration, AI-Assisted Development (Claude Code, Copilot, Cursor), Hexagonal Architecture

Experience

IBM | Software Engineering Intern | Remote

Sep 2022 – Nov 2022

- Built an internal data dashboard in React and TypeScript that pulled live team metrics from IBM REST APIs, surfacing the data across interactive charts and table views.
- Wrote TypeScript automation scripts to handle data preparation and processing tasks that fed the dashboard's visualization layer, working through implementation independently with minimal oversight.
- Delivered the project end-to-end within a 3-month timeline — from scoping and build through a live demo to management.

Projects

FinDoc Analyzer | Java 21, Spring Boot, PostgreSQL, pgvector, AWS S3/SQS, Groq, React

Apr 2026

- Architected a multi-tenant system on a ports-and-adapters (hexagonal) architecture: PostgreSQL with team_id-indexed tables and pgvector (HNSW cosine index) for team-scoped retrieval, isolation enforced at app and database layers with 58 unit tests.
- Engineered a three-stage retrieval cascade (direct vector search → LLM query rewrite → hypothetical-document fallback) using Groq (Llama 3.1 8B) and a local ONNX embedding model (all-MiniLM-L6-v2, 384-dim); parsed and chunked documents at 600 chars via the Unstructured API.
- Built 17 Spring Boot REST endpoints with JWT auth, BCrypt passwords, and three-role RBAC; used AWS SQS for async ingestion and S3 presigned URLs; 58 JUnit tests covering auth, RBAC, and tenant isolation.

Portfolio-GitOps: Zero-Touch Infrastructure | K3s, Flux CD, Infisical, OCI ARM

Feb 2026

- Architected a private K3s cluster on OCI ARM using a no-public-IP topology, eliminating direct public exposure and minimizing the cluster's attack surface end-to-end.
- Integrated Infisical with Native Kubernetes JWT authentication, replacing all static credentials with dynamic machine-identity secret injection and achieving a 100% secret-free repository state.
- Standardized a declarative GitOps lifecycle using Flux CD with automated GHCR image rollouts, cutting end-to-end infrastructure provisioning to under 10 minutes.

Instagram Sentiment Analyzer | Team Project | Python, Flask, Next.js, TypeScript, scikit-learn

May 2024

- Built a Flask REST API with 5 endpoints handling Instagram session auth, post and comment retrieval, account aggregation, and sentiment distribution; integrated a pre-trained emotion classifier into the comment pipeline.
- Developed a Next.js + TypeScript frontend with a Material-UI dashboard displaying follower stats, post metrics, and sentiment charts sourced from the Flask backend.